

HomeBased
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C450 – System Design
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<https://github.com/abaker0209/HomeBased>

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Customer Problem Statements & System Requirements

Problem Statement

Small home-based online stores depend on memory, handwritten notes, or expensive third-party solutions to manage inventory and sales, which leads to fractured data that is difficult to quantify and apply. This is especially true for stores whose inventory is often comprised of hundreds of unique items.

Customer Problem Statement

Home-based online stores are small businesses, operated from an individual's home, which typically sell handmade, or pre-owned inventory online. Their inventory is often comprised of hundreds of unique items, which are listed on multiple online listing platforms (e.g. eBay, Depop, Mercari, etc.), **HomeBased** is a web-based inventory management solution. On HomeBased, retailers can log in to quickly add new inventory items, manage existing inventory, process sales, and see insights on ways to improve their business. An individual visits the site, logs in and is greeted by a home page with suggested tasks. The user selects "add new inventory" and is taken to the "new inventory" page. The page prompts the user for information specific to the item being added, the user enters the requested information, then clicks "complete". The item is added to the "unlisted" queue. Once the user is ready, they can navigate to the "unlisted" page and easily process new inventory listings. As items are listed, they are removed from the "unlisted" page and added to the "listed inventory" page. Users can locate items on the "listed inventory" page when they are sold. They can then easily identify where the item is stored, what platforms it needs to be removed from, and what shipping materials are needed to prepare the item for shipping.

System Requirements

No.	Priority Weight	Description
REQ-1	High	The database should have the ability to store at least 1,000 items.
REQ-2	High	The system should require people to login using a username and password.
REQ-3	High	Task initiation within the system should always occur on the core pages; home, new inventory, unlisted inventory, listed inventory, and key insights.
REQ-4	High	The system should include a navigation menu that provides users quick access to the above-mentioned pages.
REQ-5	High	The system should use forms to allow users to make changes to the database.
REQ-6	Medium	The home page should suggest tasks to the user such as; add new inventory, items to list, and process a sale.
REQ-7	High	Item categories should align with categories utilized by common listing platforms. (e.g. Collared shirt vs. shirt)

REQ-8	High	The system should allow users to easily repeat a task for multiple items without requiring the user to navigate back to the task initiation screen each time.
REQ-9	Medium	The system should allow the system to filter the inventory table by a wide range of criteria. (e.g. Item type, Price, length since first listed, etc.)
REQ-10	High	The system should allow users to quickly locate an existing item record.

Functional Requirement Specifications

Stakeholders

- 1) User
- 2) Developer

Primary Actors

- 1) User: This actor can create an account, store items records, search for, locate, and make changes to item records in their database.

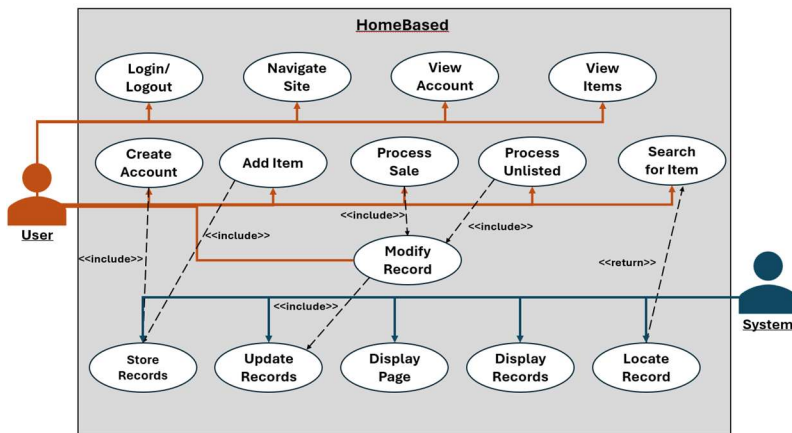
Secondary Actors

- 1) System: This is responsible for storing, updating, and providing data from the database upon instruction from the user or developer.

Use Cases

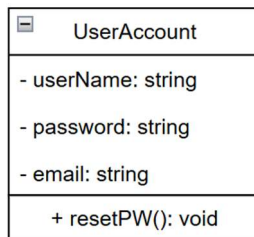
- 1) User
 - a. Create Account: To create a new user account (1)
 - b. Login/Logout: To log in or out of a personal user account (1)
 - c. Navigate Site: To quickly navigate between pages in the application. (0.5)
 - d. Add Item: To add a new item to the database (1)
 - e. Search Item: To locate an item on the database (2)
 - f. Modify Record: To modify a record on the database (2)
 - g. Process Unlisted: To access unlisted items queue and mark as listed (2)
 - h. Process Sale: To modify an item record and add a sale record to the database (2)
 - i. View Account: To view personal account details (1)
- 2) System:
 - a. Store User Details: To store user specific information within their account record. (1)
 - b. Store Item: To receive and store item details provided by the user. (1)
 - c. Locate Item: To locate a specific item matching user provided criteria. (1)
 - d. Display Items: To display all item records in the database. (1)
 - e. Display Page: To display each page available to the user on the site. (0.5)
 - f. Modify Records: To modify records within the database. (2)

Use Case Diagram

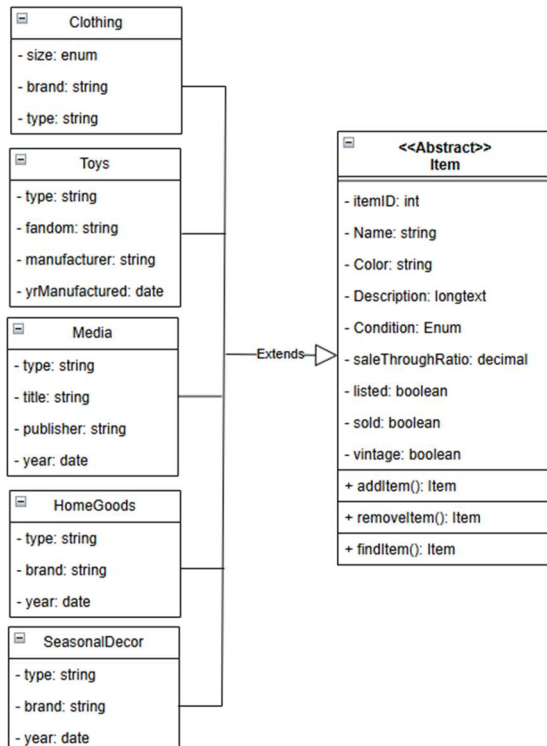


Class Diagram

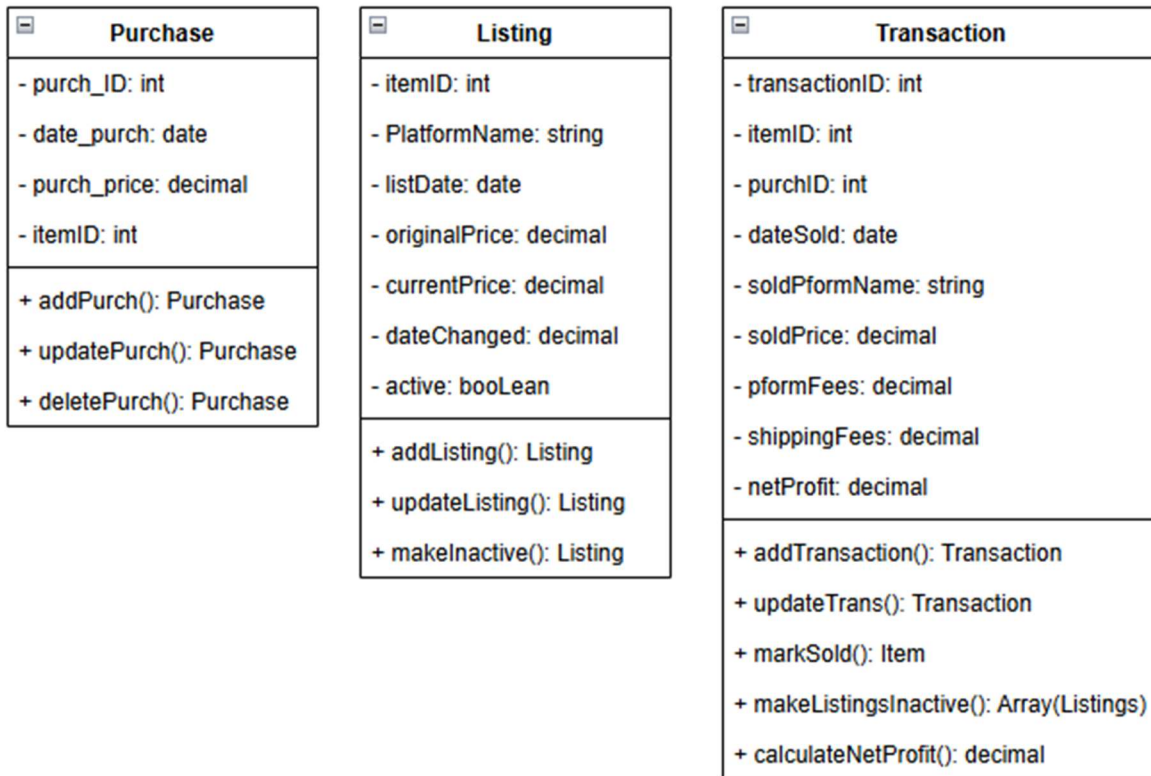
- 1) **UserAccount**: User accounts will be unique to each user and will store the username, password and email of each user. This class will also provide a function for resetting the user's password.



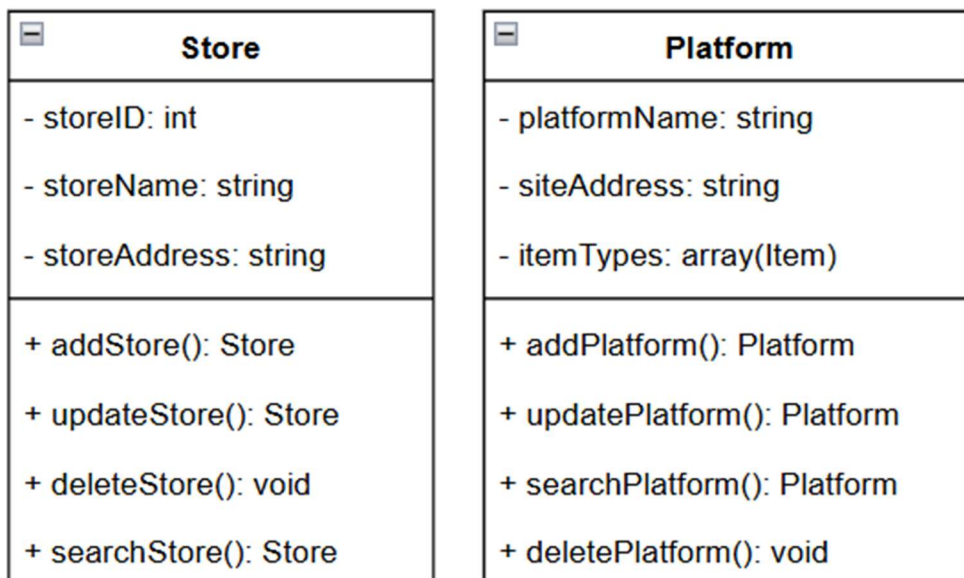
- 2) **Items**: Item records are enhanced by their type specification as the data stored for each type varies greatly. Functions within the class allow records to be added, removed and modified.



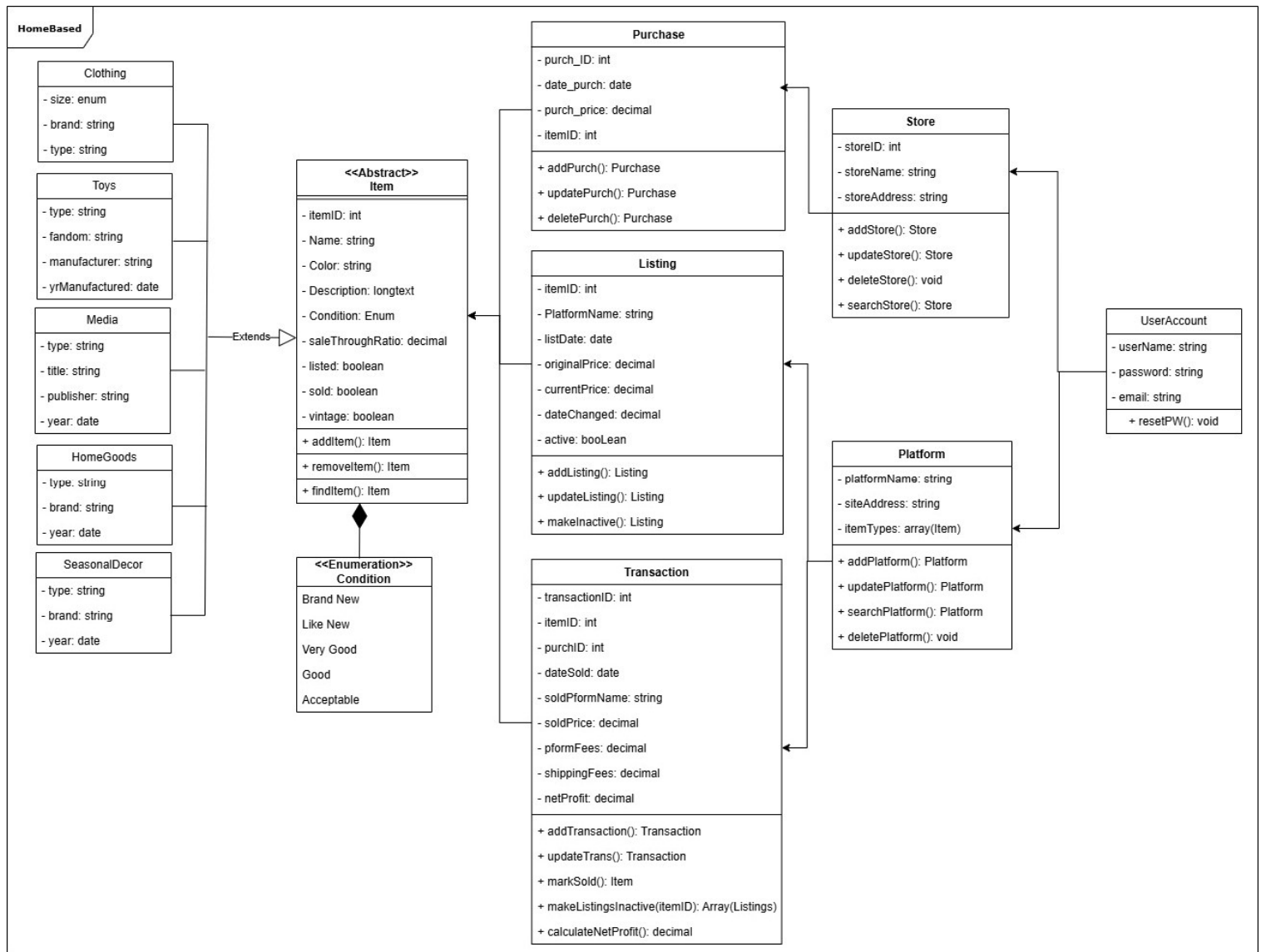
- 3) Purchase, Listing and Transaction Classes: These classes store item information specific to inventory purchases, online listings, and sales.



- 4) The store and platform classes will store information specific to the stores where inventory is being purchased, and the platforms items are listed on. These classes can be used to provide the user with key insights.



- 5) Application Composition and relationships: The user will purchase items from a store, then add them to the system. The user can then list each item on 1 or more platforms to create listing records. Listing and purchase records will create the foundation for the item records.



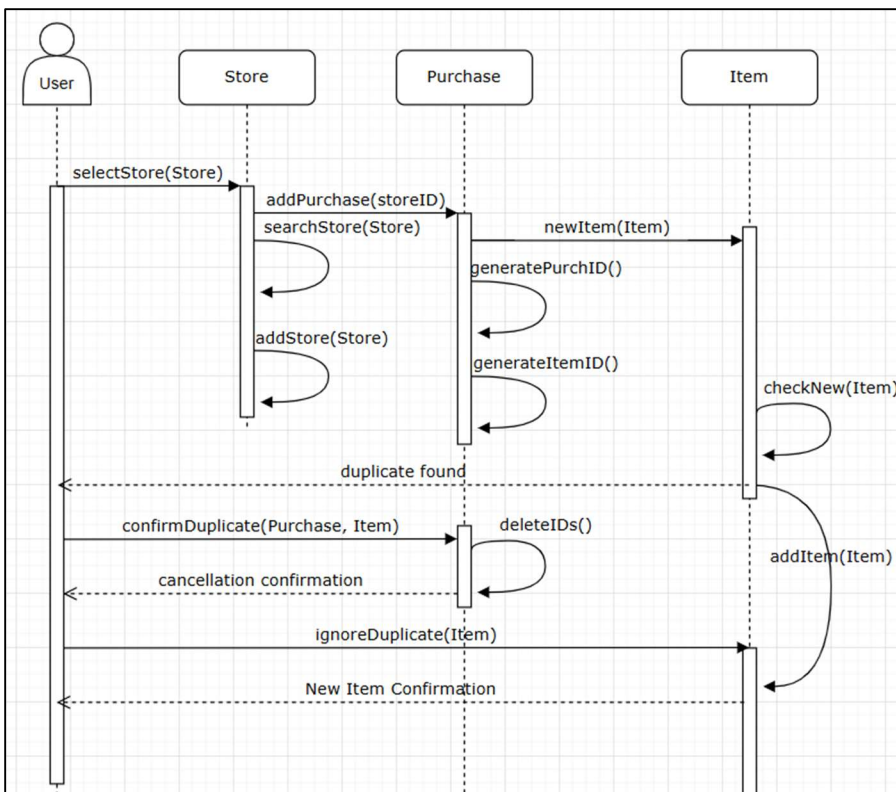
System Sequence Diagrams

Add new item

Actor: User

Objects: Store, Purchase, Item

- 1) User enters store details or selects an existing store from the dropdown menu.
- 2) Store receives store details, then compares them to existing records.
 - a. If an existing record is located, the store ID is linked to the purchase record.
 - b. If an existing record is not located, a new store ID is generated and sent to Purchase.
- 3) User enters the purchase and item details.
- 4) Purchase receives the purchase details and store ID, then generates a Purchase ID and Item ID.
- 5) Item receives the Item ID from Purchase, and the user entered item details, then compares the provided details with the existing item records.
 - a. If an existing record is located, item prompts the user to confirm if this is a duplicate entry. If the user confirms this is duplicate, the generated purchase ID and item ID are removed from the database.
 - b. If an existing record is not located, item stores the item details in the database and provides the user with a confirmation message that the action is complete.

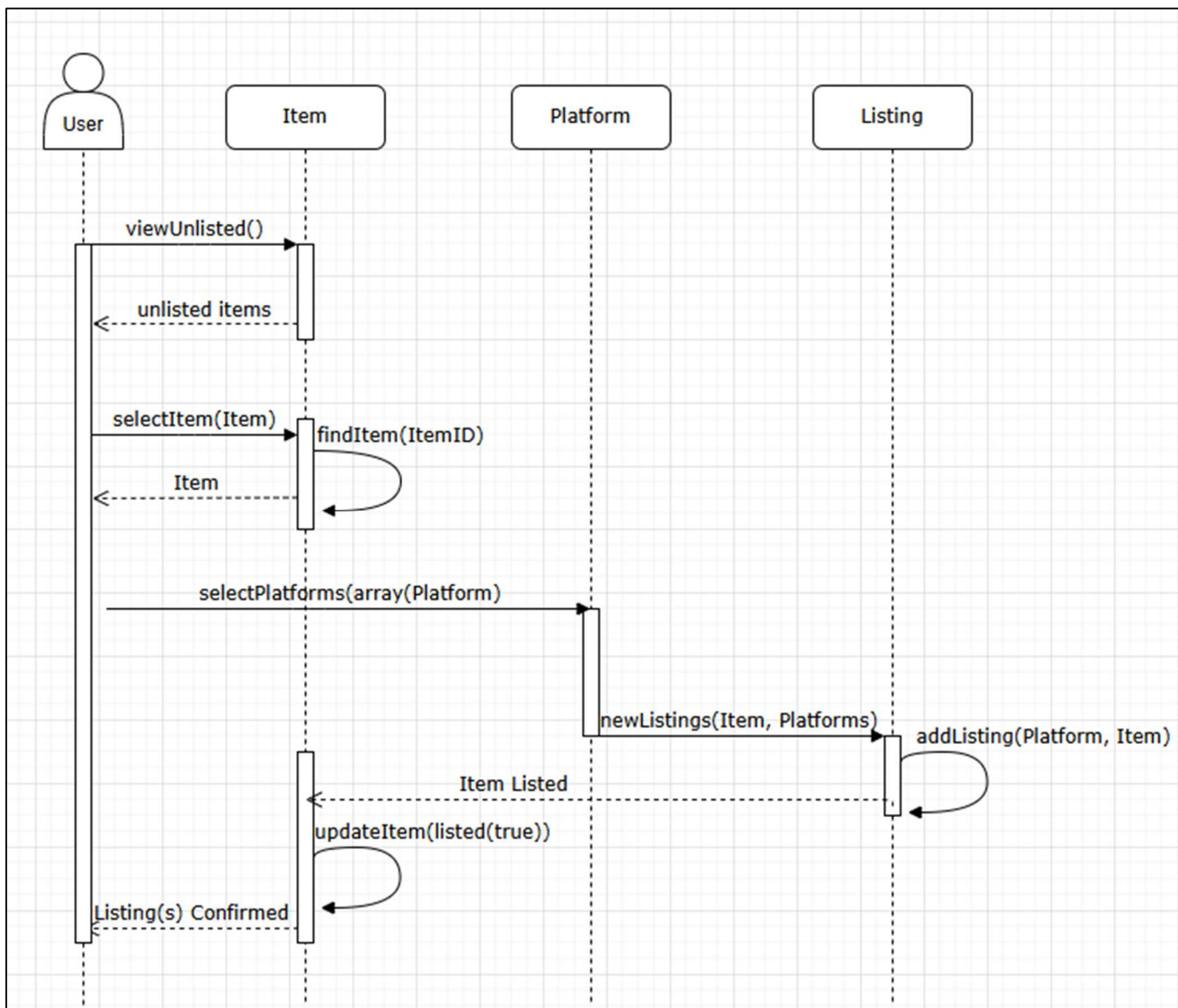


Add a new listing

Actor: User

Objects: Item, Platform, Listing

- 1) User selects to view the unlisted items queue.
- 2) Item receives the request, then returns all items with the listed variable equal to false in a FIFO queue format.
- 3) User selects an unlisted item from the queue.
- 4) Item receives the Item ID of the selected item, then returns the previously entered item details to the form.
- 5) User selects the platform(s) they wish to list the item on, enters additional details about the item, then confirms the new listing(s).
- 6) Listing receives the Item, Listing and Platform details, then generates a new Listing record for each platform selected by the user.
- 7) Item receives the additional item details, then stores the updated item record before providing the user with a confirmation message.



Activity Diagrams

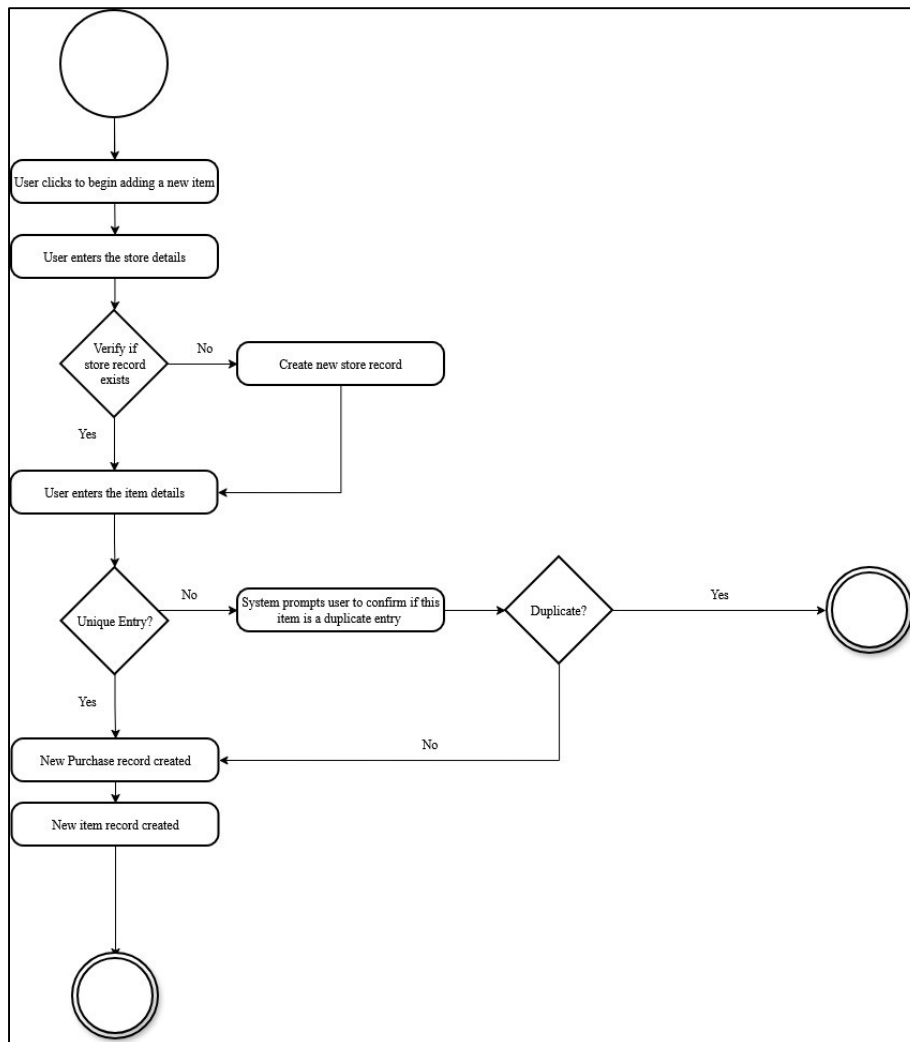
Add new item

1) States

- a) *Initial State: The user begins the process of adding a new item to the database.*
- b) *Final State:*
 - i) *The user receives confirmation that the new item was added to the database.*
 - ii) *The user cancels the action and the new item details are erased from the database.*
 - iii) *The system identifies an identical entry and notifies the user that the item is already present within the database.*

2) Actions

- a) *The user clicks a link to begin adding new items, then the system returns the new item form.*
- b) *The user enters where the item was purchased; the system connects the new item to an existing store record or creates a new store record using the information provided.*
- c) *The user enters the required item details, and the system confirms it is a unique entry before accepting the new item. Then the system prompts the user to complete the action or continue adding additional new item records.*
- d) *The user completes the action, and the system provides a confirmation screen and records the new item details within the database.*



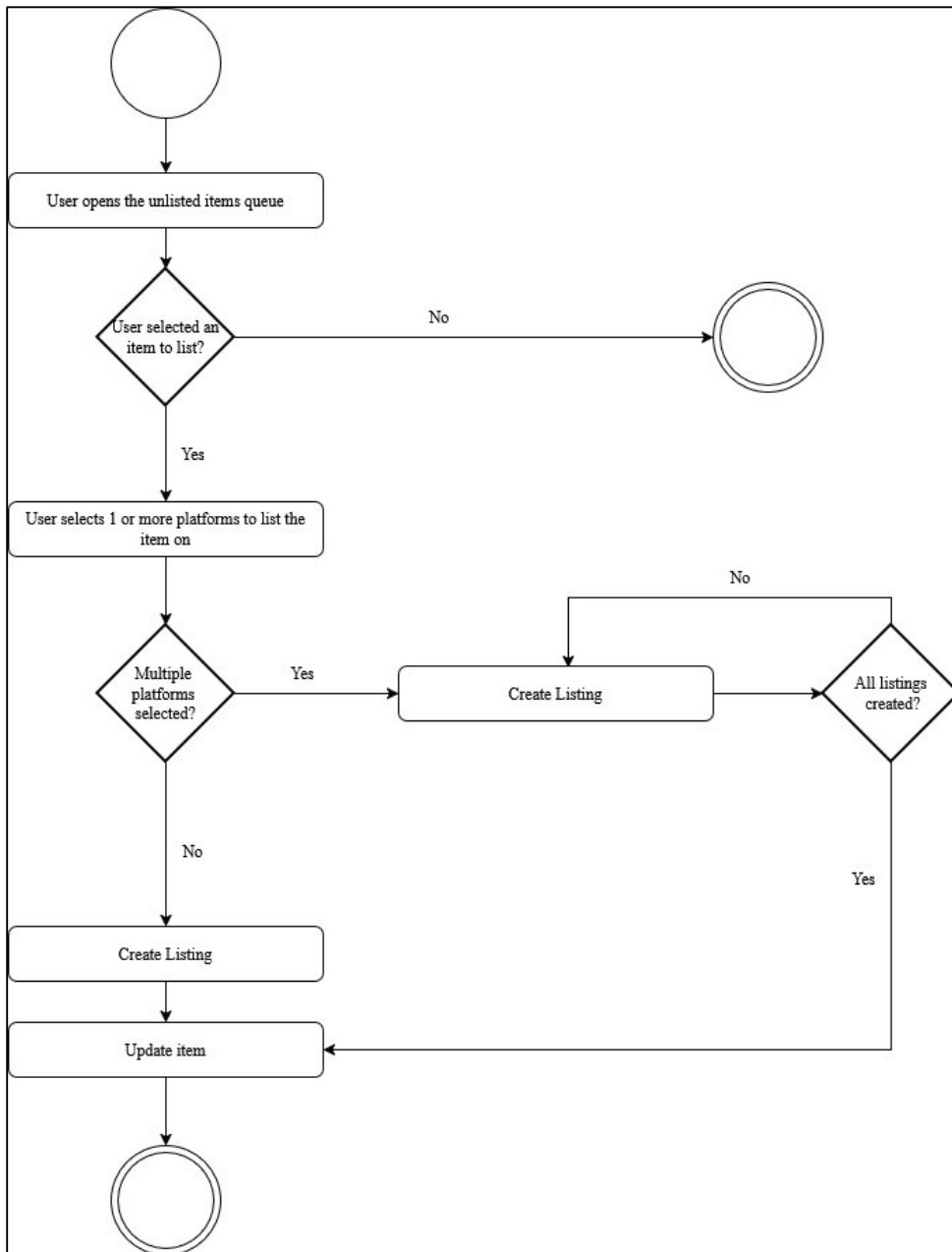
Add new Listing

1) States

- a) Initial State: The user selects an unlisted item from the queue.
- b) Final State: The system creates a new listing record and updates the item record to show that it has been listed.

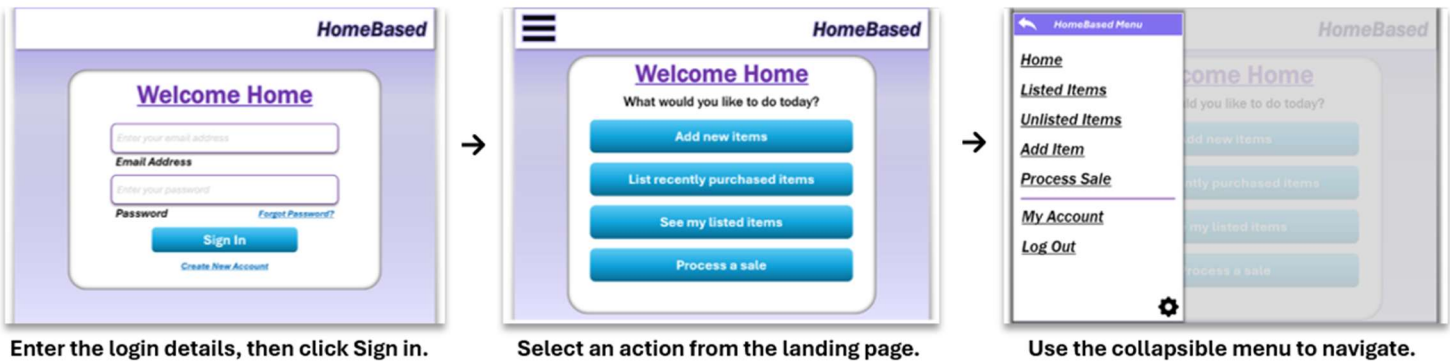
2) Actions

- a) The user opens the unlisted items queue, then selects an item to list.
- b) The user selects the platform, or platforms they wish to list the item on.
- c) The system checks if the user chose multiple platforms to list the item on.
 - i) If yes, the system iterates through each platform to create each unique listing record.
 - ii) If not, the system creates a single listing record.
- d) The system updates the item record to reflect that it has been listed.

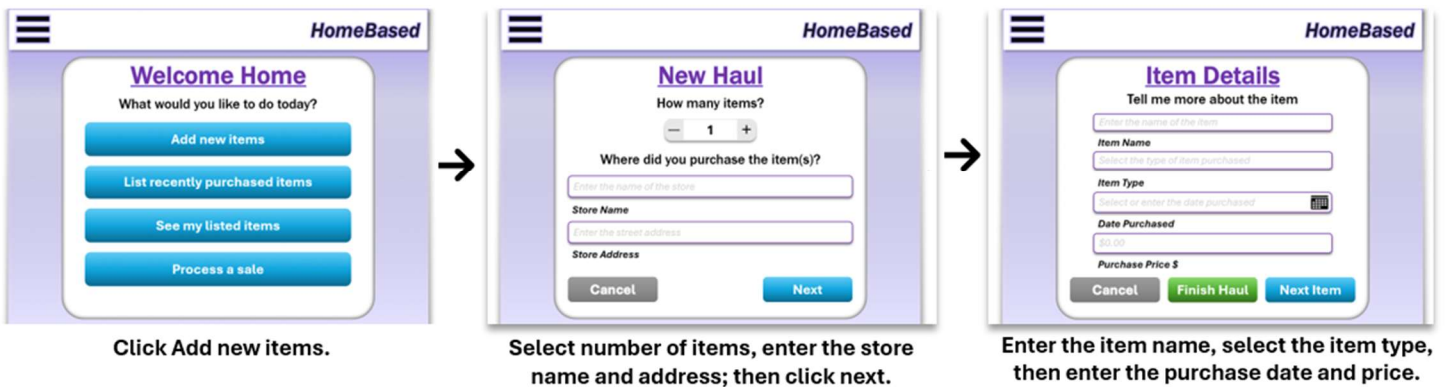


User Interface Specification

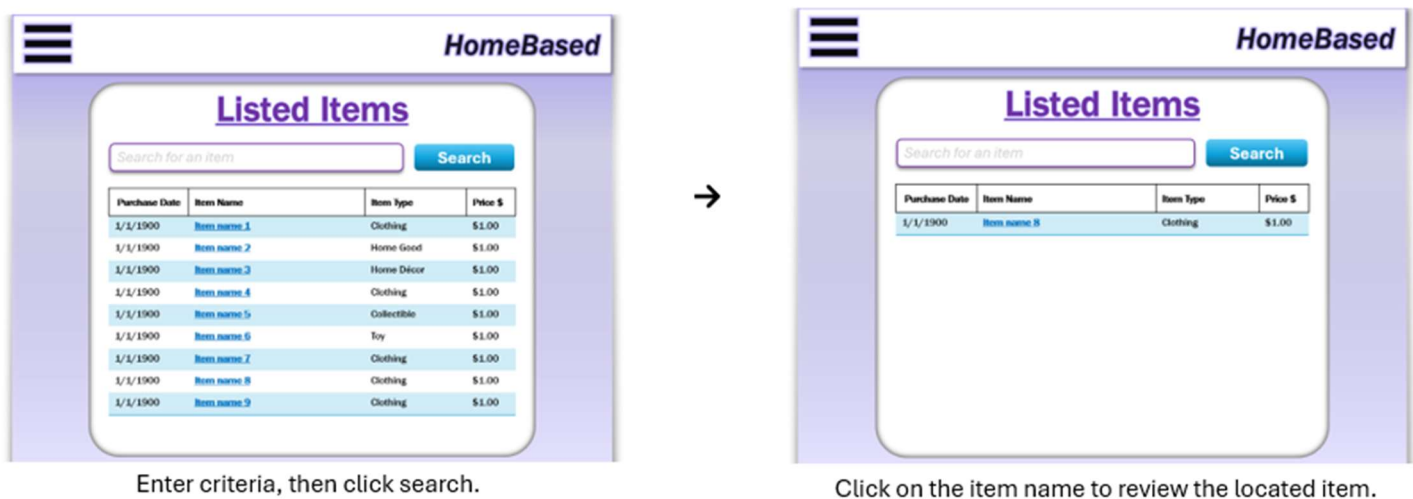
Use Case: Site Navigation



Use Case: Add a new purchase to the database



Use Case: Search for and locate an item record



Use Case: Process unlisted items within the database

HomeBased

Unlisted Items

Search for an item

Purchase Date	Item Name	Item Type	Price \$
1/1/2000	Item name 1	Clothing	\$1.00
1/1/2000	Item name 2	Household	\$1.00
1/1/2000	Item name 3	Household	\$1.00
1/1/2000	Item name 4	Clothing	\$1.00
1/1/2000	Item name 5	Clothing	\$1.00
1/1/2000	Item name 6	Toy	\$1.00
1/1/2000	Item name 7	Clothing	\$1.00
1/1/2000	Item name 8	Clothing	\$1.00
1/1/2000	Item name 9	Clothing	\$1.00

Select an item from the unlisted items queue.

HomeBased

Unlisted Item

Item Name: Date Purchased: Purchase Price: \$

What platforms will you list this item on?

eBay: ☐ Depop: ☐ Etsy: ☐ Mercari: ☐ Facebook: ☐

Cancel Next

Select which platforms to list the item on.

HomeBased

Unlisted Item

Tell me more about the item.

Is this item vintage? ☐ Yes ☐ No

Select or enter the item condition

Condition

Enter the item color

Color (if entering multiple, separate each with a comma)

Original item description

Description

Save Draft Next

Complete the form, then click next.

HomeBased

Unlisted Item

Item Type Specifics

Select or enter the item type specific detail

Specific Detail 1

Select or enter the item type specific detail

Specific Detail 2

Select or enter the item type specific detail

Specific Detail 3

Select or enter the item type specific detail

Specific Detail 4

Save Draft Next

Enter the item type specific details.

HomeBased

Unlisted Item

Tell me about recent listings for this item

Number of Active Listings: 1

Number sold in the last 6 months: 1

Enter the 1st most recent sold price \$1.00

Enter the 2nd most recent sold price \$1.00

Enter the 3rd most recent sold price \$1.00

Save Draft Skip Next

Enter details about recent listings.

HomeBased

Unlisted Item

Item Shipping Details

Item Weight lb

Height inches

Width inches

Length inches

Select the package type

Save Draft Skip Next

Enter the shipping details.

HomeBased

Unlisted Item

Item Name: Date through date: Item Type: Suggested List Price:

Listing Price: \$1.00

Click here to use the suggested list price.

Platforms this item will be listed on

Platform Platform Platform Platform

Change Attached Platforms

Save Draft Confirm Listing

Price and confirm the listing.

Use Case: Process the sale of an item within the database

HomeBased

Item Details

Item Name: Listing Date: Item Type: Original List Price:

Current Listings

Platform	Current List Price
eBay	\$1.00
Depop	\$1.00
Mercari	\$1.00
Poshmark	\$1.00

Cancel Mark Sold Edit Listing

After locating the item and opening the record, click mark sold.

HomeBased

Sold Item Details

Tell me about this sale

Platform Sold On eBay

Sold Price \$1.00

Click here if sold for the full list price.

Fees charged by the Platform \$0.00

Cancel Next

Select the platform, enter the sold price and platform fees, then click next.

HomeBased

Sold Item Details

Let's Prepare this item to ship

Estimated Shipping Costs \$0.00

Whose paying for shipping? ☐ Buyer ☐ Seller

Suggested Package Package Type 1

Item Storage Location Storage Location

Cancel Next

Enter the shipping information, locate the item and click next.

HomeBased

Sold Item Details

Item Name	Item Name 1	Sold Price	\$1.00
Item Type	Clothing	Platform Fees	\$1.50
Platform	eBay	Shipping Fees	\$0.25

Estimated Net Profit \$0.25

The below active Listings will become inactive:

Platform	Current List Price
Depop	\$1.00
Mercari	\$1.00
Poshmark	\$1.00

Cancel Confirm

Review the details of the sale, then click confirm.

HomeBased

You sold an item. Congrats!

Estimated Net Profit \$0.25

Item Name	Item Name 1	Sold Price	\$1.00
Item Type	Clothing	Platform Fees	\$1.50
Platform	eBay	Shipping Fees	\$0.25

Home

Confirmation screen.

Project Plan

- 1) Database Design & Testing (Complete)
- 2) Complete Add Item Forms / Add test items to database
- 3) Build View Unlisted Items Page
 - a. Build Item Card to view existing item records
- 4) Build Listed Items Page
 - a. Build specialized Item Card with additional details
- 5) Build Dashboard & Home Page
 - a. Dashboard with visual representations of existing data
 - b. Home Page with key insights & links to primary application actions
- 6) Build Login Pages & User Validation Functions
- 7) Deploy on AWS
 - a. Ongoing Testing
 - b. Plan for second iteration using user feedback